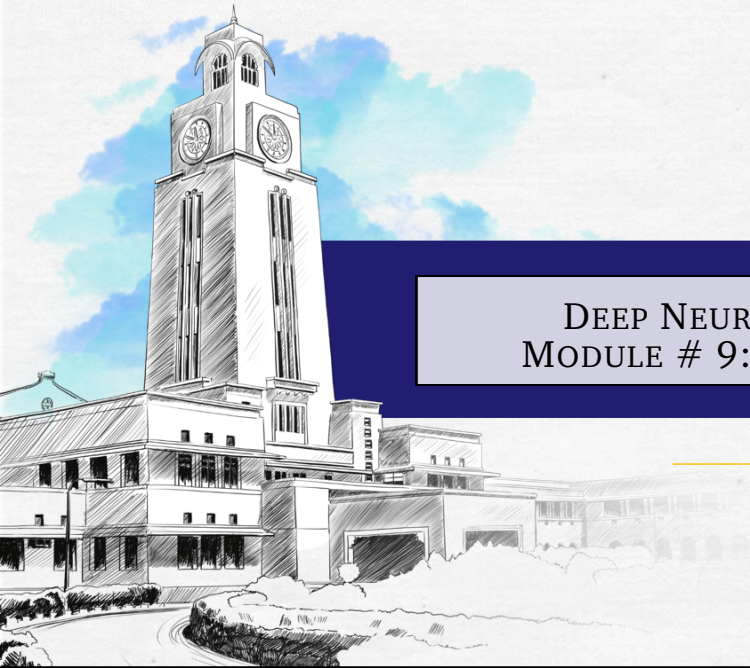




BITS Pilani
Pilani | Dubai | Goa | Hyderabad

DEEP NEURAL NETWORK MODULE # 9: TRANSFORMER

Seetha Parameswaran
BITS Pilani WILP

The instructor is gratefully acknowledging
the authors who made their course
materials freely available online.
This deck is prepared by Seetha Parameswaran.

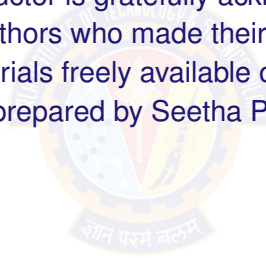


TABLE OF CONTENTS

- 1 MOTIVATION AND INDUSTRY APPLICATIONS
- 2 TRANSFORMER ARCHITECTURE
- 3 ENCODER ARCHITECTURE
- 4 DECODER ARCHITECTURE
- 5 VISION TRANSFORMER
- 6 TRANSFORMER VARIANTS
- 7 USING PRE-TRAINED MODELS



RECAP: ATTENTION MECHANISMS

- What We Learned:

- ▶ Attention allows models to focus on relevant parts of input
- ▶ Query-Key-Value framework: Attention as information retrieval
- ▶ Multiple attention types:
 - ★ Self-attention: Q, K, V from same sequence
 - ★ Multi-head: Parallel attention with different projections
 - ★ Cross-attention: Q from one sequence, K, V from another
 - ★ Masked: Prevent attending to future tokens

- What's Missing?

- ▶ How to organize these attention mechanisms into a complete architecture?
- ▶ How to handle sequences without recurrence?
- ▶ How to scale to billions of parameters?

- This Module:

- ▶ Transformer architecture: Attention as the only mechanism
- ▶ Three variants: Encoder-only, Decoder-only, Encoder-decoder

THE RNN BOTTLENECK

- Recurrent Neural Networks (RNNs):
 - ▶ Process sequences one token at a time: $h_t = f(h_{t-1}, x_t)$
 - ▶ Hidden state carries information forward
 - ▶ Used with attention
- Three Critical Problems:
 - ▶ **Sequential Processing:** Cannot parallelize
 - ★ Must compute h_1 before h_2 before h_3 ...
 - ★ Training is slow (hours $\rightarrow$ days $\rightarrow$ weeks)
 - ▶ **Vanishing Gradients:** Long-range dependencies fail
 - ★ Information from early tokens fades
 - ★ Even with LSTM/GRU, struggles beyond 100-200 tokens
 - ▶ **Fixed Memory:** Hidden state is bottleneck
 - ★ All information compressed into fixed-size vector
 - ★ Limited capacity as sequences grow
- Question: Can we build sequence models without recurrence?

THE TRANSFORMER SOLUTION (2017)

- Key Insight: "Attention is All You Need"
 - ▶ Replace recurrence entirely with self-attention
 - ▶ Each token directly attends to all other tokens
 - ▶ No sequential dependency → full parallelization

- How Transformers Solve RNN Problems:

Problem	RNN	Transformer
Parallelization	Sequential	Parallel
Long-range deps	Vanishing gradients	Direct connections
Memory	Fixed hidden state	Attention to all tokens

- The Impact:
 - ▶ 2017: Machine translation breakthrough
 - ▶ 2018: BERT revolutionizes NLP (encoder-only)
 - ▶ 2018-2020: GPT series (decoder-only)
 - ▶ 2020-present: Foundation of all large language models
- Result: Transformers became universal architecture for sequences

FROM TRANSFORMERS TO MODERN AI

- Transformers Enable Scaling:
 - ▶ Parallel processing → train on massive datasets
 - ▶ Direct connections → learn complex patterns
 - ▶ Flexible architecture → adapt to any domain
- This Scaling Led To:
 - ① **Large Language Models (LLMs)**
 - ★ 100B+ parameters: GPT-4, Claude, Gemini
 - ★ Trained on trillions of tokens
 - ★ General intelligence: reasoning, coding, writing
 - ② **Small Language Models (SLMs)**
 - ★ 1B-7B parameters: Phi-3, Gemma, Mistral-7B
 - ★ On-device deployment: phones, laptops
 - ★ Privacy-preserving, low-latency
 - ③ **Specialized Models**
 - ★ Vision: ViT, CLIP (images without convolutions)
 - ★ Speech: Whisper, Wav2Vec2 (audio understanding)
 - ★ Multimodal: GPT-4V, Gemini (text + images + audio)

REAL-WORLD APPLICATIONS

① Conversational AI

- ▶ ChatGPT, Claude, Gemini: Natural dialogue with context
- ▶ Customer support automation: 24/7 intelligent responses
- ▶ Virtual assistants: Multi-turn conversations with memory

② Code Generation & Software Development

- ▶ GitHub Copilot: Real-time code completion
- ▶ CodeLlama, StarCoder: Function generation from descriptions

③ Content Creation

- ▶ Writing assistance: Articles, emails, reports
- ▶ Summarization: Long documents → key points
- ▶ Translation: 100+ languages with context

④ Scientific Research

- ▶ AlphaFold 3: Protein structure prediction
- ▶ Drug discovery: Molecular design and screening

⑤ Next: How do these models differ? Three transformer variants

THREE TRANSFORMER ARCHITECTURES

- Why Three Variants?

- ▶ Different tasks need different information flow
- ▶ Trade-off between understanding vs generation
- ▶ Computational efficiency for specific use cases

Encoder-Only

Bidirectional

Attention



Classification

Decoder-Only

Causal

Attention



Generation

Encoder-Decoder

Encoder

→ Decoder



Seq2Seq

- Selection Criteria:

- ▶ Need to classify or extract? → Encoder-only
- ▶ Need to generate text? → Decoder-only
- ▶ Input and output very different? → Encoder-decoder

INDUSTRY MODELS COMPARISON

Model	Arch	Parameters	Strengths	Access
GPT-4	Decoder	1.8T	Reasoning, code	API (OpenAI)
Claude 4	Decoder	200B	Safety, long context	API (Anthropic)
Gemini 3	Decoder	175B	Multimodal	API (Google)
LLaMA 3	Decoder	8B-405B	Open, efficient	Open (Meta)
Mistral	Decoder	7B-123B	Fast inference	Open (Mistral AI)
BERT	Encoder	110M-340M	Understanding, NER	Open (Google)
T5	Enc-Dec	60M-11B	Translation, summary	Open (Google)

- Key Observations:

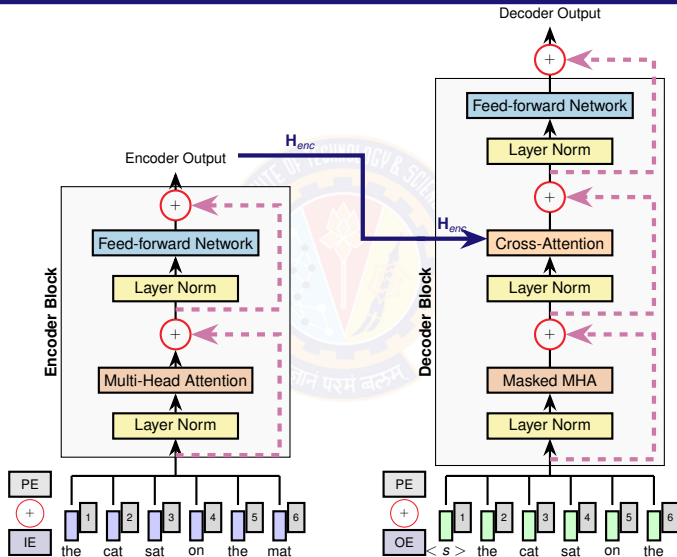
- ▶ Decoder-only dominates generative tasks
- ▶ Open models (LLaMA, Mistral) competitive with proprietary
- ▶ Parameter count vs performance: Not always linear

TABLE OF CONTENTS

- 1 MOTIVATION AND INDUSTRY APPLICATIONS
- 2 TRANSFORMER ARCHITECTURE
- 3 ENCODER ARCHITECTURE
- 4 DECODER ARCHITECTURE
- 5 VISION TRANSFORMER
- 6 TRANSFORMER VARIANTS
- 7 USING PRE-TRAINED MODELS



TRANSFORMER ARCHITECTURE



WHAT DOES THE ENCODER DO?

- **Purpose:** Transform input sequence into rich contextual representations
 - ▶ Input: Raw tokens (words, subwords, patches)
 - ▶ Output: Context-aware embeddings for each input position
- **Key Capability: Bidirectional Context**
 - ▶ Each token attends to **all other tokens** (no restrictions)
 - ▶ Captures relationships in both directions
 - ▶ Example: "bank" in "river bank" vs "bank account"
- **How it Works:**
 - ▶ Multi-Head Attention: Tokens "communicate" with each other
 - ▶ Feed-Forward Network: Each token "thinks" independently
 - ▶ Stacked N times: Builds increasingly abstract representations
- **Output:**
 - ▶ $\mathbf{H}_{enc} \in \mathbb{R}^{n \times d_{model}}$ where n is sequence length
 - ▶ Each row is a contextualized representation of one input token
 - ▶ Used by decoder (encoder-decoder) or task head (encoder-only)

WHAT DOES THE DECODER DO?

- **Purpose:** Generate output sequence one token at a time
 - ▶ Input: Previously generated tokens (autoregressive)
 - ▶ Output: Probability distribution over next token
- **Key Capability: Causal Generation**
 - ▶ Token at position t can only attend to positions $\leq t$
 - ▶ Prevents "looking ahead" to future tokens
 - ▶ Ensures model learns to predict sequentially
- **How it Works:**
 - ▶ Masked Self-Attention: Tokens attend only to past
 - ▶ Cross-Attention: Attends to encoder output (encoder-decoder only)
 - ▶ Feed-Forward Network: Process information independently
 - ▶ Stacked N times: Refines generation at each layer
- **Output:**
 - ▶ $\mathbf{D}_{out} \in \mathbb{R}^{m \times d_{model}}$ where m is target length
 - ▶ Select next token (greedy/sampling/beam search)

TABLE OF CONTENTS

- 1 MOTIVATION AND INDUSTRY APPLICATIONS
- 2 TRANSFORMER ARCHITECTURE
- 3 ENCODER ARCHITECTURE**
- 4 DECODER ARCHITECTURE
- 5 VISION TRANSFORMER
- 6 TRANSFORMER VARIANTS
- 7 USING PRE-TRAINED MODELS



TRANSFORMER ENCODER: INPUT EMBEDDING

- Input Embedding:

- ▶ Convert tokens to dense vectors:

$$\mathbf{x}_i \in \mathbb{R}^{d_{\text{model}}}$$

- ▶ Learned embedding matrix: $\mathbf{E} \in \mathbb{R}^{|V| \times d_{\text{model}}}$

- ▶ Typical $d_{\text{model}} = 512$ (base) or 1024 (large)



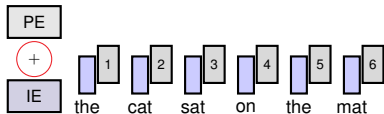
TRANSFORMER ENCODER: POSITION EMBEDDING

- Positional Encoding:

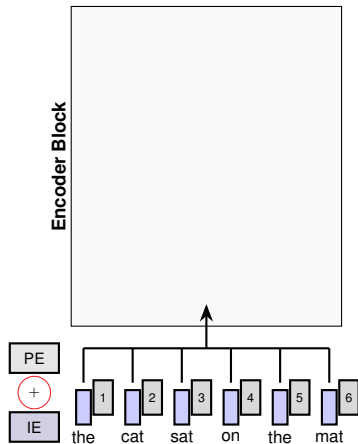
- ▶ Problem: Attention has no notion of position
- ▶ Solution: Add position information to embeddings

$$PE(pos, 2i) = \sin\left(\frac{pos}{10000^{2i/d_{model}}}\right) \quad (1)$$

$$PE(pos, 2i + 1) = \cos\left(\frac{pos}{10000^{2i/d_{model}}}\right) \quad (2)$$

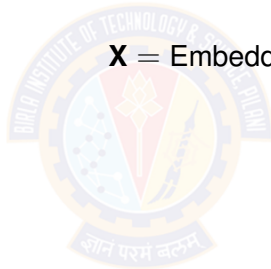


TRANSFORMER ENCODER: ENCODER INPUT

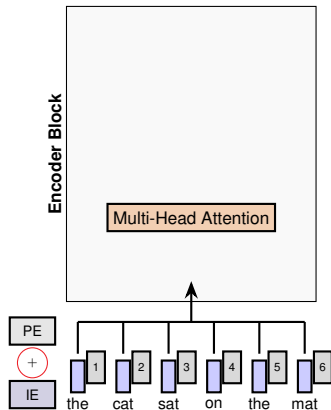


- **Final Encoder Input:** = Input Embedding concatenate Position Embedding

$$\mathbf{X} = \text{Embedding}(\text{tokens}) + \text{PE}(\text{positions}) \quad (3)$$



TRANSFORMER ENCODER: MHA



Multi-Head Attention:

- Allows each embedding to attend to all other embeddings
- h parallel attention heads capture different relationships
- **For each head** $h = 1, \dots, H$:

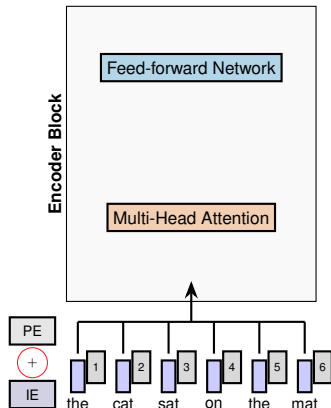
$$\mathbf{Q}_h = \mathbf{Z}\mathbf{W}_h^Q, \quad \mathbf{K}_h = \mathbf{Z}\mathbf{W}_h^K, \quad \mathbf{V}_h = \mathbf{Z}\mathbf{W}_h^V \quad (4)$$

$$\text{head}_h = \text{softmax} \left(\frac{\mathbf{Q}_h \mathbf{K}_h^T}{\sqrt{d_k}} \right) \mathbf{V}_h \quad (5)$$

$$\text{MHA}(\mathbf{Z}) = \text{Concat}(\text{head}_1, \dots, \text{head}_H) \mathbf{W}^O \quad (6)$$

- **Typical:** $H = 12$ heads, $d_k = d_v = D/H = 64$

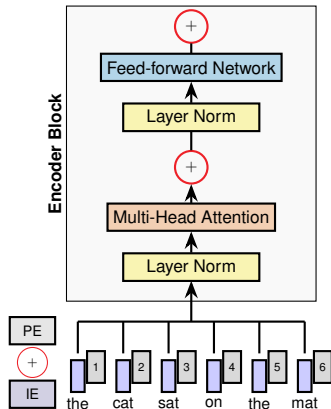
TRANSFORMER ENCODER: FFN



Feed-Forward Network (Position-wise):

- FFN "Thinks" **independently** and processes the information
- Two-layer FFN with expansion
 - ▶ **Expand:** $D \rightarrow 4D$; $\mathbf{W}_1 \in \mathbb{R}^{D \times 4D}$ parameters
 - ▶ **Non-linearity:** RELU activation
 - ▶ Applied **independently** to each position
 - ▶ **Contract:** $4D \rightarrow D$; $\mathbf{W}_2 \in \mathbb{R}^{4D \times D}$ parameters
- Why Position-wise FFN?
 - ▶ Same weights applied at each position independently
 - ▶ Adds non-linearity after attention
 - ▶ Provides model capacity for complex transformations

TRANSFORMER ENCODER: LAYER NORM



• Problem: Training Deep Networks is Hard

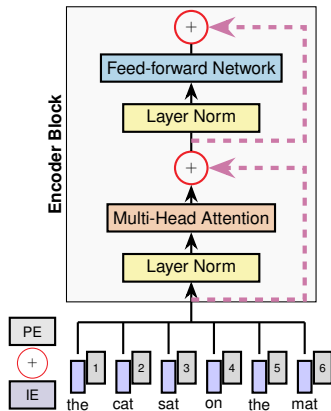
- ▶ Stacking many layers → vanishing gradients
- ▶ Information loss
- ▶ Optimization becomes unstable

• Solution 1: Layer Normalization

- ▶ Normalize across feature dimension
- ▶ Stabilizes training across features (not batch)
- ▶ Learnable parameters: $\gamma, \beta \in \mathbb{R}^{d_{\text{model}}}$
- ▶ Same procedure at train/test time

$$\text{LN}(\mathbf{x}) = \gamma \frac{\mathbf{x} - \mathbb{E}[\mathbf{x}]}{\sqrt{\text{Var}[\mathbf{x}] + \epsilon}} + \beta \quad (7)$$

TRANSFORMER ENCODER: RESIDUAL CONNECTION



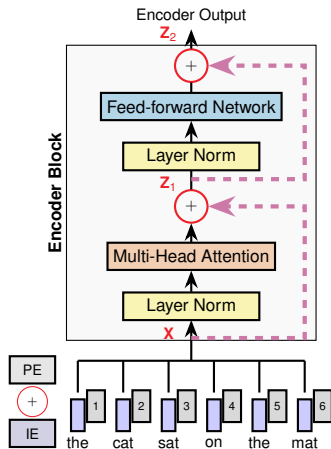
- Problem: Training Deep Networks is Hard
- Solution 2: Residual Connections (Skip Connections)

- ▶ Help gradient flow in deep networks (from ResNet)
- ▶ **Add input directly to output:**

$$\text{Out} = \mathbf{x} + \text{Sublayer}(\mathbf{x}) \quad (8)$$

- ▶ Creates "gradient highways" → easier optimization
- ▶ Allows training very deep models (100+ layers)

TRANSFORMER ENCODER: ENCODER OUTPUT



- Complete Encoder Layer:

$$Z_1 = X + \text{MHA}(\text{LayerNorm}(X)) \quad (9)$$

$$Z_2 = Z_1 + \text{FFN}(\text{LayerNorm}(Z_1)) \quad (10)$$

TRANSFORMER ENCODER: PARAMETERS

Each token gets embedded into a d_{model} -dimensional vector, and we can process sequences up to n_{max} tokens long.

- $|V|$: Vocabulary size (30K-50K tokens)
- d_{model} : Embedding vector length/dimension (512 (base) or 1024 (large))
- n_{max} : Maximum sequence length (512-2048 tokens)

Input Embedding: $|V| \times d_{\text{model}}$

Positional Encoding: $n_{\text{max}} \times d_{\text{model}}$ (learnable)

Layer Norm (2 layers): $2 \times 2 \times d_{\text{model}} = 4d_{\text{model}}$

Multi-Head Attention: $4 \times d_{\text{model}}^2$ (for W^Q, W^K, W^V, W^O)

Feed-Forward Network: $2 \times d_{\text{model}} \times d_{\text{ff}} = 8d_{\text{model}}^2$ where $d_{\text{ff}} = 4d_{\text{model}}$

$$W_1 \in \mathbb{R}^{d_{\text{model}} \times d_{\text{ff}}}, \quad W_2 \in \mathbb{R}^{d_{\text{ff}} \times d_{\text{model}}}$$

Total per layer: $4d_{\text{model}} + 4d_{\text{model}}^2 + 8d_{\text{model}}^2 \approx 12d_{\text{model}}^2$

ENCODER STACK

- Stack of N Identical Layers:
 - ▶ Original Transformer: $N = 6$ layers
 - ▶ BERT-base: $N = 12$ layers
 - ▶ BERT-large: $N = 24$ layers
 - ▶ GPT-3: $N = 96$ layers (decoder, but same principle)

- Stacking Process:

$$\mathbf{H}^{(\ell)} = \text{EncoderLayer}^{(\ell)}(\mathbf{H}^{(\ell-1)}), \quad \ell = 1, \dots, N \quad (11)$$

where $\mathbf{H}^{(0)} = \text{InputEmbedding} + \text{PositionalEncoding}$

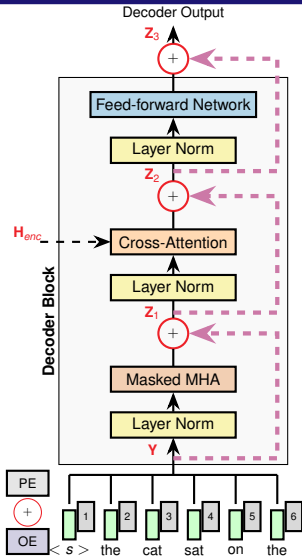
- Encoder Output:
 - ▶ Final encoder output: $\mathbf{H}^{(N)} \in \mathbb{R}^{n \times d_{\text{model}}}$
 - ▶ Rich contextual representation for each input token
 - ▶ Each position has attended to all other positions N times
 - ▶ Passed to decoder via cross-attention (in encoder-decoder models)

TABLE OF CONTENTS

- 1 MOTIVATION AND INDUSTRY APPLICATIONS
- 2 TRANSFORMER ARCHITECTURE
- 3 ENCODER ARCHITECTURE
- 4 DECODER ARCHITECTURE**
- 5 VISION TRANSFORMER
- 6 TRANSFORMER VARIANTS
- 7 USING PRE-TRAINED MODELS



TRANSFORMER DECODER: DECODER OUTPUT



- Complete Decoder Layer:

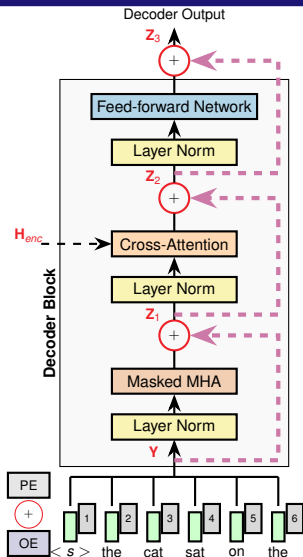
$$\mathbf{Z}_1 = \mathbf{Y} + \text{MaskedMHA}(\text{LayerNorm}(\mathbf{Y})) \quad (12)$$

$$\mathbf{Z}_2 = \mathbf{Z}_1 + \text{CrossAttn}(\text{LayerNorm}(\mathbf{Z}_1), \mathbf{H}_{enc}) \quad (13)$$

$$\mathbf{Z}_3 = \mathbf{Z}_2 + \text{FFN}(\text{LayerNorm}(\mathbf{Z}_2)) \quad (14)$$

where $\mathbf{H}_{enc}$ is encoder output

TRANSFORMER DECODER: MASKED MULTI-HEAD ATTENTION

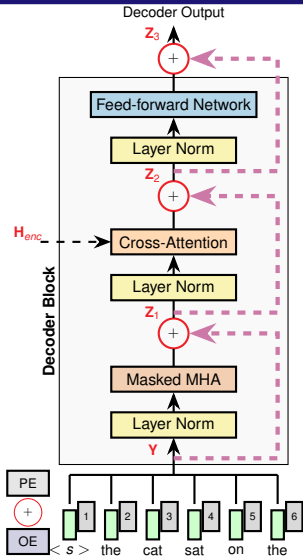


- Self-attention with causal masking
 - ▶ Token t attends only to positions $\leq t$
 - ▶ Prevents "looking ahead" to future tokens
- **Why?** Match training and inference
 - ▶ At inference: Generate one token at a time
 - ▶ Training must simulate this autoregressive process

$$\text{mask}_{ij} = \begin{cases} 0 & \text{if } i \geq j \\ -\infty & \text{if } i < j \end{cases} \quad (15)$$

$$\text{MaskedAttn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax} \left(\frac{\mathbf{QK}^T}{\sqrt{d_k}} + \text{mask} \right) \mathbf{V} \quad (16)$$

TRANSFORMER DECODER: CROSS-ATTENTION



Cross-Attention (Encoder-Decoder Attention):

- Decoder attends to encoder output
 - ▶ Query (Q): From decoder layer
 - ▶ Key (K), Value (V): From encoder output H_{enc}
 - ▶ No masking (can attend to all source tokens)
- **Why needed:** Connect source and target
 - ▶ Allows decoder to focus on relevant source positions

$$\text{CrossAttn}(\mathbf{Q}_{dec}, \mathbf{K}_{enc}, \mathbf{V}_{enc}) = \text{softmax} \left(\frac{\mathbf{Q}_{dec} \mathbf{K}_{enc}^T}{\sqrt{d_k}} \right) \mathbf{V}_{enc}$$

$$\mathbf{Q}_{dec} = \text{LN}(\mathbf{Z}_1) \mathbf{W}^Q$$

$$\mathbf{K}_{enc} = \mathbf{H}_{enc} \mathbf{W}^K$$

$$\mathbf{V}_{enc} = \mathbf{H}_{enc} \mathbf{W}^V$$

TRANSFORMER DECODER: PARAMETERS

Each output token gets embedded into a d_{model} -dimensional vector.

- Same vocabulary and embedding dimensions as encoder
- $|V|$, d_{model} , n_{max} same as encoder

Output Embedding: $|V| \times d_{\text{model}}$

Positional Encoding: $n_{\text{max}} \times d_{\text{model}}$ (learnable)

Layer Norm (3 layers): $3 \times 2 \times d_{\text{model}} = 6d_{\text{model}}$

Masked Multi-Head Attention: $4 \times d_{\text{model}}^2$ (for W^Q, W^K, W^V, W^O)

Cross-Attention: $4 \times d_{\text{model}}^2$ (for W^Q, W^K, W^V, W^O)

Feed-Forward Network: $2 \times d_{\text{model}} \times d_{\text{ff}} = 8d_{\text{model}}^2$ where $d_{\text{ff}} = 4d_{\text{model}}$

$$W_1 \in \mathbb{R}^{d_{\text{model}} \times d_{\text{ff}}}, \quad W_2 \in \mathbb{R}^{d_{\text{ff}} \times d_{\text{model}}}$$

Total per layer: $6d_{\text{model}} + 8d_{\text{model}}^2 + 8d_{\text{model}}^2 \approx 16d_{\text{model}}^2$

Note: Decoder has 33% more parameters than encoder (16 vs 12 d_{model}^2)

DECODER STACK

- Stack of N Identical Decoder Layers:
 - ▶ Original Transformer: $N = 6$ layers
 - ▶ Each layer has **3 sublayers** (vs 2 in encoder):
 - 1 Masked multi-head self-attention
 - 2 Cross-attention to encoder output
 - 3 Position-wise feed-forward network
- Stacking Process:

$$\mathbf{D}^{(\ell)} = \text{DecoderLayer}^{(\ell)}(\mathbf{D}^{(\ell-1)}, \mathbf{H}_{enc}), \quad \ell = 1, \dots, N \quad (17)$$

where $\mathbf{D}^{(0)} = \text{OutputEmbedding} + \text{PositionalEncoding}$

- Decoder Output:
 - ▶ Final decoder output: $\mathbf{D}^{(N)} \in \mathbb{R}^{m \times d_{\text{model}}}$ where m is target length
 - ▶ Each position has attended to previous positions N times (masked)
 - ▶ Each position has attended to all encoder positions N times (cross-attention)

OUTPUT LAYER: LINEAR + SOFTMAX

- **Purpose:** Convert decoder output to vocabulary probabilities
- **Linear Projection:**

$$\text{logits} = \mathbf{D}^{(N)} \mathbf{W}_{\text{out}} + \mathbf{b}_{\text{out}} \quad (18)$$

where $\mathbf{W}_{\text{out}} \in \mathbb{R}^{d_{\text{model}} \times |V|}$, $\mathbf{b}_{\text{out}} \in \mathbb{R}^{|V|}$

- **Softmax:** Output: Probability distribution over vocabulary

$$P(y_t | y_{<t}, \mathbf{x}) = \text{softmax}(\text{logits}_t) \quad (19)$$

- **Complete Flow:**

$$\mathbb{R}^{d_{\text{model}}} \xrightarrow{\text{Linear}} \mathbb{R}^{|V|} \xrightarrow{\text{Softmax}} \text{Probabilities} \xrightarrow{\text{Token Selection}} y_t$$

- Token selection done via greedy, sampling, or beam search

TABLE OF CONTENTS

- 1 MOTIVATION AND INDUSTRY APPLICATIONS
- 2 TRANSFORMER ARCHITECTURE
- 3 ENCODER ARCHITECTURE
- 4 DECODER ARCHITECTURE
- 5 VISION TRANSFORMER**
- 6 TRANSFORMER VARIANTS
- 7 USING PRE-TRAINED MODELS



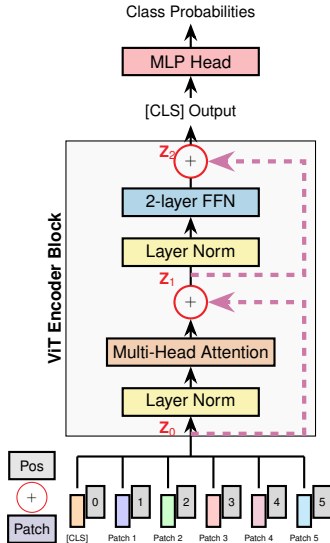
FROM NLP TO VISION - WHY ViTs WORK

- Traditional Approach: CNNs
 - ▶ Convolutional layers extract local features
 - ▶ Pooling layers for spatial hierarchy
 - ▶ Inductive bias: Translation equivariance, locality
- The ViT (Vision Transformer) Innovation (2020):
 - ▶ Treat image as sequence of patches
 - ▶ Apply pure transformer (no convolutions)
 - ▶ Learn visual relationships through self-attention
- Key Insight:
 - ▶ CNNs: Hard-coded locality through convolution kernel
 - ▶ ViTs: Learn relationships between all patches
 - ▶ Trade-off: Need more data, but more flexible
- When ViTs Excel:
 - ▶ Large-scale pretraining (ImageNet-21K, JFT-300M)
 - ▶ Transfer learning to downstream tasks

WHAT DOES VISION TRANSFORMER (ViT) Do?

- **Purpose:** Image classification using pure transformer (no convolutions)
- **Architecture:** Encoder-only (no decoder needed for classification)
- **Input → Output:**
- **Key Innovation:** Treat image as sequence of patches
 - ▶ Divide image into fixed-size patches (16×16)
 - ▶ Flatten each patch → Linear projection
 - ▶ Add positional encoding
 - ▶ Process through transformer encoder
- **How Classification Works:**
 - ▶ Prepend [CLS] token to patch sequence
 - ▶ Transformer encoder processes all patches
 - ▶ .[CLS] output → MLP head → Class probabilities
- **Why It Works:** Self-attention captures spatial relationships globally

VISION TRANSFORMER (ViT): ARCHITECTURE



- **Architecture:** Encoder-only (no decoder)
 - ▶ Input: Image $\rightarrow$ Output: Class probabilities
 - ▶ Uses only transformer encoder blocks
 - ▶ No decoder needed for classification task
- **Classification [CLS] Token:**
 - ▶ Learnable special token prepended to patch sequence
 - ▶ Aggregates global image information via self-attention
 - ▶ Final [CLS] representation used for classification
- **Processing Flow:**
Image $\rightarrow$ Patches $\rightarrow$ Encoder $\rightarrow$ [CLS] $\rightarrow$ Class

PATCH EMBEDDING & [CLS] TOKEN

- Image to Patches:

- ▶ Input image: $\mathbf{I} \in \mathbb{R}^{H \times W \times C}$ (e.g., $224 \times 224 \times 3$)
- ▶ Divide into N patches of size $P \times P$ (typically $P = 16$)
- ▶ Number of patches: $N = \frac{H \times W}{P^2} = \frac{224 \times 224}{16 \times 16} = 196$

- Linear Projection:

$$\mathbf{z}_i = \text{Flatten}(\text{patch}_i) W_p, \quad W_p \in \mathbb{R}^{(P^2 \cdot C) \times d} \quad (20)$$

- ▶ Flatten each patch: $16 \times 16 \times 3 = 768$ pixels
- ▶ Project to embedding dimension d (typically 768)
- ▶ Result: $\mathbf{Z} = [\mathbf{z}_1; \mathbf{z}_2; \dots; \mathbf{z}_N] \in \mathbb{R}^{N \times d}$

- Prepend [CLS] Token:

- ▶ Learnable embedding $\mathbf{z}_{[\text{CLS}]} \in \mathbb{R}^d$ prepended to sequence
- ▶ Aggregates global image information via self-attention
- ▶ Final sequence: $[\mathbf{z}_{[\text{CLS}]}; \mathbf{z}_1; \mathbf{z}_2; \dots; \mathbf{z}_N] \in \mathbb{R}^{(N+1) \times d}$

POSITIONAL ENCODING

- **Problem:** Self-attention is permutation-invariant
 - ▶ Model cannot distinguish patch positions without explicit encoding
 - ▶ Order of patches matters for understanding image structure
- **Solution: Learnable Positional Embeddings**
 - ▶ $\mathbf{E}_{\text{pos}} \in \mathbb{R}^{(N+1) \times d}$ where $N + 1 = 197$ positions
 - ▶ One embedding for each position: [CLS] (position 0) + 196 patches
 - ▶ Learned during training (not fixed like sinusoidal in transformers)
 - ▶
 - ▶ 1D embeddings sufficient; no 2D structure hardcoded
- **Adding Position Information:**

$$\mathbf{z}_0^{(0)} = [\mathbf{z}_{[\text{CLS}]}; \mathbf{z}_1; \mathbf{z}_2; \dots; \mathbf{z}_N] + \mathbf{E}_{\text{pos}} \quad (21)$$

- ▶ Element-wise addition of position embeddings
- ▶ Input to first encoder layer: $\mathbf{z}_0^{(0)} \in \mathbb{R}^{(N+1) \times d}$

ViT ENCODER: PROCESSING

- Encoder Layer Operations:

- ▶ Same as Transformer encoder:

Layer Norm $\rightarrow$ MHA $\rightarrow$ Add $\rightarrow$ Layer Norm $\rightarrow$ FFN $\rightarrow$ Add

- ▶ Stack of L identical layers (ViT-Base: $L = 12$)

- Layer ℓ aligns: $\mathbf{z}_0^{(\ell-1)} \in \mathbb{R}^{(N+1) \times d}$ is input to layer ℓ

$$\mathbf{z}_1^{(\ell)} = \mathbf{z}_0^{(\ell-1)} + \text{MHA}(\text{LayerNorm}(\mathbf{z}_0^{(\ell-1)})) \quad (22)$$

$$\mathbf{z}_2^{(\ell)} = \mathbf{z}_1^{(\ell)} + \text{FFN}(\text{LayerNorm}(\mathbf{z}_1^{(\ell)})) \quad (23)$$

- Multi-Head Self-Attention:

- ▶ Each patch (including [CLS]) attends to all other patches
- ▶ Captures global spatial relationships
- ▶ ViT-Base: $H = 12$ heads, $d_k = d/H = 64$

- Feed-Forward Network: $W_1 \in \mathbb{R}^{d \times 4d}$, $W_2 \in \mathbb{R}^{4d \times d}$ (expansion factor 4)

$$\text{FFN}(\mathbf{x}) = \text{GELU}(\mathbf{x}W_1 + b_1)W_2 + b_2 \quad (24)$$

- Gaussian Error Linear Unit $\text{GELU}(x) = x \cdot \Phi(x)$ where $\Phi(x)$ is the cumulative distribution function (CDF) of the standard normal distribution.

ViT OUTPUT: CLASSIFICATION

- Final Encoder Output:

- ▶ After L layers: $\mathbf{z}_2^{(L)} \in \mathbb{R}^{(N+1) \times d}$
- ▶ Extract [CLS] token representation: $\mathbf{z}_{[\text{CLS}]}^{(L)} \in \mathbb{R}^d$
- ▶ This contains aggregated information from entire image

- Classification Head (MLP):

$$\mathbf{h} = \text{LayerNorm}(\mathbf{z}_{[\text{CLS}]}^{(L)}) \quad (25)$$

$$\text{logits} = \mathbf{h} \mathbf{W}_{\text{head}} + \mathbf{b}_{\text{head}} \quad (26)$$

where $\mathbf{W}_{\text{head}} \in \mathbb{R}^{d \times K}$, K is number of classes

- Softmax for Probabilities:

$$P(y = k | \mathbf{I}) = \text{softmax}(\text{logits})_k \quad (27)$$

- ▶ Output: Probability distribution over K classes
- ▶ Final prediction: $\hat{y} = \arg \max_k P(y = k | \mathbf{I})$

- Training: Cross-entropy loss between predictions and true labels

ViT vs CNNs - TRADE-OFFS

Aspect	CNNs	ViT
Inductive Bias	Strong (locality, translation equivariance)	Weak (learns from data)
Data Efficiency	Good with small datasets	Requires large-scale pretraining
Scalability	Limited by kernel size	Scales well with model size
Global Context	Limited (small receptive field)	Excellent (self-attention)
Compute	Efficient convolutions	Quadratic in sequence length
Transfer Learning	Good	Excellent (better than CNNs)

- Practical Guidelines:

- ▶ **Use CNNs:** Small datasets ($<100K$ images), limited compute
- ▶ **Use ViT:** Large-scale pretraining available, transfer learning
- ▶ **Hybrid:** Combine CNN stem with Transformer body

TABLE OF CONTENTS

- 1 MOTIVATION AND INDUSTRY APPLICATIONS
- 2 TRANSFORMER ARCHITECTURE
- 3 ENCODER ARCHITECTURE
- 4 DECODER ARCHITECTURE
- 5 VISION TRANSFORMER
- 6 TRANSFORMER VARIANTS**
- 7 USING PRE-TRAINED MODELS



WHY THREE TRANSFORMER VARIANTS?

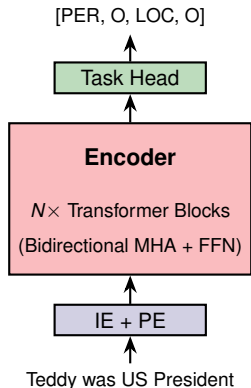
- Different Tasks Need Different Architectures
 - ▶ Not all NLP tasks require generation
 - ▶ Not all tasks need bidirectional context
 - ▶ Trade-offs between capability and efficiency
- Key Design Decision: Attention Pattern

Aspect	Encoder-Only	Decoder-Only	Encoder-Decoder
Attention	Bidirectional	Causal (masked)	Both
Information Flow	All tokens see all	Token sees only past	Encoder: bi, Decoder: causal
Task Type	Understanding	Generation	Transformation
Use Case	Full context needed	Sequential output	Input $\neq$ Output
Efficiency	Single pass, parallel	Simpler, scalable	Most flexible, compute-intensive

ENCODER-ONLY ARCHITECTURE

- Architecture:
 - ▶ Only encoder stack (no decoder)
 - ▶ Bidirectional self-attention: Each token attends to all tokens
 - ▶ Output: Rich contextual representations for each input token
- Pretraining: Masked Language Modeling (MLM)
 - ▶ Randomly mask 15% of input tokens
 - ▶ Predict masked tokens using context from both directions
 - ▶ Example: "The [MASK] sat on the mat" → predict "cat"
- Use Cases:
 - ▶ Text classification: Sentiment analysis, spam detection
 - ▶ Named Entity Recognition (NER): Extract names, locations
 - ▶ Question Answering: Extract answer spans from context
 - ▶ Sentence embeddings: Semantic similarity, clustering

TRANSFORMER: ENCODER-ONLY ARCHITECTURE



- Popular Models:

- ▶ BERT (2018): 110M-340M parameters
- ▶ RoBERTa (2019): Optimized BERT training
- ▶ DistilBERT (2019): 40% smaller, 60% faster

- Encoder-Only (Named Entity Recognition):

- ▶ Input 1: "Teddy is a soft toy" → Output: [O, O, O, O, O] (O=other)
- ▶ Input 2: "Teddy was US President" → Output: [PER, O, LOC, O]
- ▶ Processes entire sentence at once (bidirectional context)

DECODER-ONLY ARCHITECTURE

- Architecture:

- ▶ Only decoder stack (no encoder)
- ▶ Causal/masked self-attention: Token t attends only to $\leq t$
- ▶ Output: Probability distribution over next token

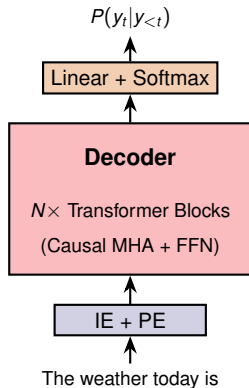
- Pretraining: Autoregressive Language Modeling

- ▶ Predict next token given previous tokens
- ▶ Training objective: $P(x_t | x_{<t})$
- ▶ Example: "The cat sat on" → predict "the"

- Use Cases:

- ▶ Text generation: Creative writing, content creation
- ▶ Conversational AI: ChatGPT, Claude, Gemini
- ▶ Code generation: GitHub Copilot, CodeLlama
- ▶ Few-shot learning: In-context learning without fine-tuning

TRANSFORMER: DECODER-ONLY ARCHITECTURE



- Popular Models:

- ▶ GPT-4 (2023): 1.8T parameters (estimated)
- ▶ Claude 4 (2025): 200B parameters
- ▶ LLaMA 3 (2024): 8B-405B parameters (open)
- ▶ Mistral (2023): 7B-123B parameters (open)

- Decoder-Only (Text Completion):

- ▶ Prompt: "The weather today is"
- ▶ Generates: "sunny and warm with a gentle breeze"
- ▶ Each token generated based on all previous tokens

ENCODER-DECODER ARCHITECTURE

- Architecture:

- ▶ Full transformer: Both encoder and decoder stacks
- ▶ Encoder: Bidirectional self-attention on input
- ▶ Decoder: Causal self-attention + cross-attention to encoder

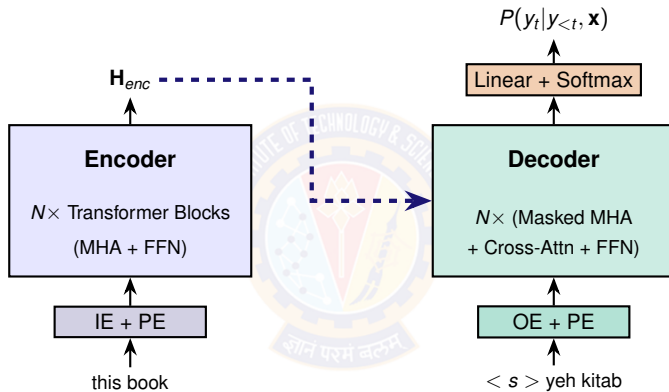
- Pretraining: Various Objectives

- ▶ T5: Text-to-text format (all tasks as generation)
- ▶ BART: Denoising autoencoder (corrupt and reconstruct)
- ▶ mT5: Multilingual text-to-text

- Use Cases:

- ▶ Machine translation: English → French, etc.
- ▶ Summarization: Long document → concise summary
- ▶ Question answering: Question + context → answer
- ▶ Data-to-text: Structured data → natural language

ENCODER-DECODER TRANSFORMER



Popular Models:

- T5 (2019): 60M-11B parameters
- mT5 (2020): Multilingual T5 (101 languages)

ENCODER-DECODER (TRANSLATION) EXAMPLE

- Training:

- ▶ Input: "this book is good" → Target: "yeh kitab acchi hai"
- ▶ Decoder sees all target tokens with masking
- ▶ Predicts all tokens in parallel (teacher forcing)

- Inference:

- ▶ Step 1: ["<start>"] → "yeh"
- ▶ Step 2: ["<start>", "yeh"] → "kitab" (continues...)

COMPARISON: WHEN TO USE WHAT

Aspect	Encoder-Only	Decoder-Only	Encoder-Decoder
Attention	Bidirectional	Causal	Both
Use Case	Understanding	Generation	Transformation
Tasks	Classification, NER, QA	Chat, completion	Translation, summary
Examples	Sentiment, entity tagging	Code gen, dialogue	Eng→Hindi, abstractive
Models	BERT, RoBERTa	GPT-4, LLaMA	T5, BART, mT5
Context	Full sentence	Left-to-right	Source: full, Target: causal
Speed	Fast (parallel)	Slow (sequential)	Medium

Selection Guide:

- Need to classify or extract? → Encoder-only
- Need to generate text? → Decoder-only
- Input language $\neq$ Output language? → Encoder-decoder

LARGE-SCALE PRETRAINING PARADIGM

- Why Pretrain?

- ▶ Learn general language understanding from massive unlabeled data
- ▶ Transfer knowledge to downstream tasks with limited labeled data
- ▶ Better performance than training from scratch

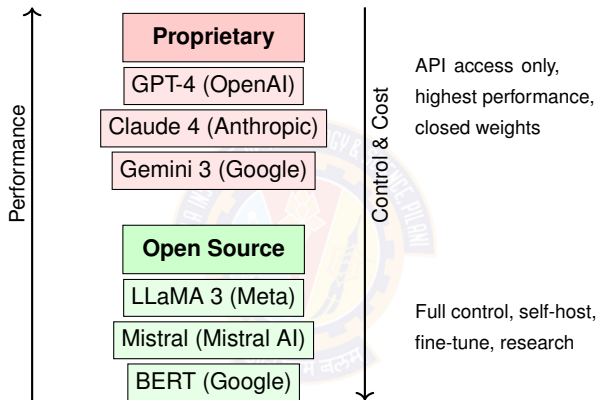
- Pretraining Objectives by Architecture:

- ▶ **Encoder-only:** Masked Language Modeling (BERT, RoBERTa)
- ▶ **Decoder-only:** Autoregressive LM (GPT, LLaMA)
- ▶ **Encoder-decoder:** Span corruption, denoising (T5, BART)

- Fine-tuning Approaches:

- ▶ Full fine-tuning: Update all parameters
- ▶ Parameter-efficient: LoRA, adapters (update $< 1\%$ params)
- ▶ Prompting: Zero-shot, few-shot (no parameter updates)

FOUNDATION MODELS LANDSCAPE



- **Trade-offs:**

- ▶ **Proprietary:** Best performance, no control, API costs
- ▶ **Open:** Full control, hosting costs, requires expertise

TABLE OF CONTENTS

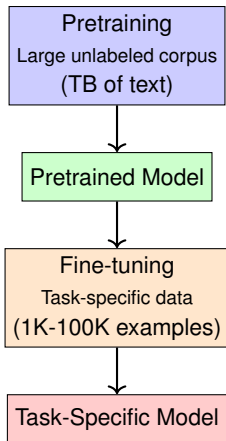
- 1 MOTIVATION AND INDUSTRY APPLICATIONS
- 2 TRANSFORMER ARCHITECTURE
- 3 ENCODER ARCHITECTURE
- 4 DECODER ARCHITECTURE
- 5 VISION TRANSFORMER
- 6 TRANSFORMER VARIANTS
- 7 USING PRE-TRAINED MODELS



- What is HuggingFace?
 - ▶ Open platform for ML models, datasets, and applications
 - ▶ 500,000+ pre-trained models available
 - ▶ Easy-to-use transformers library (Python)
 - ▶ Community-driven: researchers, companies, hobbyists
- Popular Models on HuggingFace:
 - ▶ **Encoder-only:** bert-base-uncased, roberta-large, distilbert-base
 - ▶ **Decoder-only:** gpt2, meta-llama/Llama-3-8B, mistralai/Mistral-7B
 - ▶ **Encoder-decoder:** t5-base, facebook/bart-large, google/flan-t5-large
- Benefits: Instant access, no training needed, proven performance

TRANSFER LEARNING & FINE-TUNING

Transfer Learning Paradigm:



Why Not Train From Scratch?

- Requires massive data (billions of tokens)
- Expensive compute (thousands of GPU-hours)
- Pretrained models already learned general language patterns

Self-supervised: MLM,
autoregressive

Supervised: Classification,
NER, etc.

FINE-TUNING WORKFLOW

- Step 1: Load Pretrained Model
- Step 2: Prepare Task-Specific Data
 - ▶ Tokenize inputs with pretrained tokenizer
 - ▶ Create train/validation splits
 - ▶ Prepare labels in correct format
- Step 3: Fine-tune with Small Learning Rate
 - ▶ Typical LR: 2×10^{-5} to 5×10^{-5}
 - ▶ Few epochs: 2-5 (avoid overfitting)
 - ▶ Use Adam optimizer with weight decay
- Step 4: Evaluate on Validation Set
 - ▶ Monitor metrics: Accuracy, F1, loss
 - ▶ Early stopping if validation performance degrades

FINE-TUNING EXAMPLE: SENTIMENT ANALYSIS

- Task: Classify movie reviews as positive/negative
- Dataset: IMDb (25K train, 25K test)
- Base Model: BERT or RoBERTa

```
1 from transformers import (AutoTokenizer, Trainer,  
2     AutoModelForSequenceClassification, TrainingArguments)  
3  
4 # Load model and tokenizer  
5 tokenizer = AutoTokenizer.from_pretrained('bert-base-uncased')  
6 model = AutoModelForSequenceClassification.from_pretrained(  
7     'bert-base-uncased', num_labels=2  
8 )  
9  
10 # Tokenize data  
11 def tokenize_function(examples):  
12     return tokenizer(examples['text'], padding='max_length',  
13                     truncation=True, max_length=512)  
14  
15 tokenized_train = train_dataset.map(tokenize_function, batched=True)  
16 tokenized_val = val_dataset.map(tokenize_function, batched=True)
```

FINE-TUNING EXAMPLE: SENTIMENT ANALYSIS

```
1  # Training arguments
2  training_args = TrainingArguments(
3      output_dir='./results', num_train_epochs=3,
4      per_device_train_batch_size=16, per_device_eval_batch_size=64,
5      learning_rate=2e-5, weight_decay=0.01,
6      evaluation_strategy='epoch', save_strategy='epoch',
7      load_best_model_at_end=True,
8  )
9
10 # Trainer
11 trainer = Trainer(model=model, args=training_args,
12                   train_dataset=tokenized_train, eval_dataset=tokenized_val,
13 )
14
15 # Train
16 trainer.train()
17
18 # Evaluate
19 results = trainer.evaluate()
20 print(f"Accuracy: {results['eval_accuracy']:.4f}")
```

BEST PRACTICES FOR FINE-TUNING

- Learning Rate:
 - ▶ Use small learning rates: 2×10^{-5} to 5×10^{-5}
- Layer Freezing Strategies:
 - ▶ **Full fine-tuning**: Update all parameters (best performance)
 - ▶ **Gradual unfreezing**: Start with top layers, unfreeze gradually
 - ▶ **Freeze embeddings**: Only update task-specific layers
- Hyperparameter Selection:
 - ▶ Batch size: 16-32 (balance memory and convergence)
 - ▶ Epochs: 2-5 (more epochs → overfitting risk)
 - ▶ Weight decay: 0.01-0.1 (regularization)
 - ▶ Gradient clipping: Max norm 1.0 (stability)
- Data Considerations:
 - ▶ Need at least 1,000 labeled examples for fine-tuning
 - ▶ More data (10K-100K) → better performance

EVALUATION METRICS

- Classification Tasks:
 - ▶ **Accuracy:** $\frac{\text{Correct predictions}}{\text{Total predictions}}$
 - ▶ **F1 Score:** Harmonic mean of precision and recall
 - ▶ **Confusion Matrix:** Detailed breakdown of predictions
- Named Entity Recognition (NER):
 - ▶ **Entity-level F1:** Precision and recall on full entities
 - ▶ **Token-level accuracy:** Per-token classification accuracy
- Question Answering:
 - ▶ **Exact Match (EM):** Percentage of exact answer matches
 - ▶ **F1 Score:** Token overlap between prediction and ground truth
- Text Generation:
 - ▶ **BLEU:** Precision-based metric for translation/generation
 - ▶ **ROUGE:** Recall-based metric for summarization
 - ▶ **Perplexity:** How well model predicts text

PARAMETER-EFFICIENT FINE-TUNING

- **Problem:** Full fine-tuning updates all parameters
 - ▶ BERT-base: 110M parameters
 - ▶ Memory and compute intensive
- **LoRA (Low-Rank Adaptation):**
 - ▶ Freeze pretrained weights
 - ▶ Add trainable low-rank matrices: $\Delta W = BA$
 - ▶ $B \in \mathbb{R}^{d \times r}, A \in \mathbb{R}^{r \times d}$ where $r \ll d$
 - ▶ Update only $< 1\%$ of parameters

$$W_{\text{new}} = W_{\text{frozen}} + \alpha \cdot BA \quad (28)$$

- **Benefits:**
 - ▶ 10-100× fewer trainable parameters
 - ▶ Lower memory footprint
 - ▶ Faster training
 - ▶ Comparable or better performance

SUMMARY

- What We Learned:

- ▶ Transformers revolutionized AI through pure attention
- ▶ Three architectures: Encoder-only, Decoder-only, Encoder-decoder
- ▶ Foundation of all modern LLMs and multimodal models
- ▶ HuggingFace makes pretrained models easily accessible
- ▶ Fine-tuning adapts models to specific tasks efficiently

- Emerging Trends:

- ▶ Multimodal Transformers: Text + Image + Audio (GPT-4V, Gemini)
- ▶ Efficiency Improvements: Sparse attention, linear attention, LoRA
- ▶ Long Context: 100K-1M token context windows (Gemini, Claude)
- ▶ Mixture of Experts (MoE): Sparse activation for efficiency

- Practical Takeaway:

- ▶ Start with pretrained models from HuggingFace
- ▶ Fine-tune on your task-specific data
- ▶ Use parameter-efficient methods for large models

REFERENCES

- Zhang, A., Lipton, Z. C., Li, M., & Smola, A. J. (2023). *Dive into deep learning*. Cambridge University Press. Chapters 11.
- Goodfellow, I., Bengio, Y., Courville, A. (2016). *Deep Learning*. MIT Press.